# Sea Water Pressure Signal Malfunction

## Symptoms

- The OEM sea water pressure sensor has malfunctioned.

## How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

## Shoptalk

The sea water pressure sensor is connected to the Alarm and Safety C4 connector located on the customer interface box.

# Troubleshooting Summary

| STEPS | | SPECIFICATIONS |
|---|---|---|
| **STEP 1.** | Check the customer interface box wiring. | |
| | **STEP 1A.** Check the sea water pressure signal and sensor supply +24-VDC wires for an open. | |
| | **STEP 1B.** Check the sea water pressure signal and sensor supply +24-VDC wires for a wire-to-wire short. | |
| | **STEP 1C.** Check the sea water pressure signal wire for a short to ground. | |
| **STEP 2.** | Check the OEM wiring harness. | |
| | **STEP 2A.** Check the sea water pressure signal and sensor supply +24-VDC wires for an open. | |

| STEPS | | SPECIFICATIONS |
|---|---|---|
| | **STEP 2B.** Check the sea water pressure signal and sensor supply +24-VDC wires for a wire-to-wire short. | |
| | **STEP 2C.** Check the sea water pressure signal wire for a short to ground. | |
| | **STEP 2D.** Check the sea water pressure sensor supply +24-VDC wire for voltage. | |

# STEP 1. Check the customer interface box wiring.

## STEP 1A. Check the sea water pressure signal and sensor supply +24-VDC wires for an open.

**Conditions :**

- Open the customer interface box.
- Disconnect the sea water pressure signal and sensor supply +24-VDC wires at the remote input/output unit.
- Disconnect the C4 connector.

| Action | Specification/Repair | Next Step |
|---|---|---|

<table>
<tr>
<td rowspan="2">Check the sea water pressure signal and sensor supply +24-VDC wires for an open.<br><br>NOTE: An alarm will sound on the remote input/output unit when an open is detected.<br><br>• Place one test lead on the sea water pressure signal wire at the remote input/output unit.<br>• Place the other test lead on the sea water pressure signal pin at the C4 connector.<br>• Place one test lead on the sensor supply +24-VDC wire at the remote input/output unit.<br>• Place the other test lead on the sensor supply +24-VDC pin at the C4 connector.<br><br>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</td>
<td>Less than 10 ohms?<br>YES</td>
<td>1B</td>
</tr>
<tr>
<td>Less than 10 ohms?<br>NO<br><br>Repair :<br>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</td>
<td>Repair complete</td>
</tr>
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### STEP 1B. Check the sea water pressure signal and sensor supply +24-VDC wires for a wire-to-wire short.

**Conditions :**

- Open the customer interface box.
- Disconnect the sea water pressure signal and sensor supply +24-VDC wires at the remote input/output unit.
- Disconnect the C4 connector.

| Action | Specification/Repair | Next Step |
| --- | --- | --- |

| | | |
|---|---|---|
| Check the sea water pressure signal and sensor supply +24-VDC wires for a wire-to-wire short.<br>• Place one test lead on the sea water pressure signal wire at the remote input/output unit.<br>• Place the other test lead on all other wires at the remote input/output unit.<br>• Place one test lead on the sensor supply +24-VDC wire at the remote input/output unit.<br>• Place the other test lead on all other wires at the remote input/output unit.<br>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification. | Less than 10 ohms?<br>**YES**<br>**Repair :**<br>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) | **Repair complete** |
| | Less than 10 ohms?<br>**NO** | **1C** |

### STEP 1C. Check the sea water pressure signal and sensor supply +24-VDC wires for a short to ground.

**Conditions :**

- Open the customer interface box.
- Disconnect the sea water pressure signal and sensor supply +24-VDC wires at the remote input/output unit.
- Disconnect the C4 connector.

| Action | Specification/Repair | Next Step |
|---|---|---|

| | | |
|---|---|---|
| Check the sea water pressure signal and sensor supply +24-VDC wires for a short to ground.<br>• Place one test lead on the sea water pressure signal wire at the remote input/output unit.<br>• Place the other test lead on panel ground.<br>• Place one test lead on the sensor supply +24-VDC wire at the remote input/output unit.<br>• Place the other test lead on panel ground.<br>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification. | Less than 10 ohms?<br>**YES**<br>**Repair :**<br>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) | **Repair complete** |
| | Less than 10 ohms?<br>**NO** | **2A** |

## STEP 2. Check the OEM wiring harness.

### STEP 2A. Check the sea water pressure signal and sensor supply +24-VDC wires for an open.

**Conditions :**

- Disconnect the OEM harness at the C4 and C11 connectors.
- Disconnect the sea water pressure sensor connector.

| Action | Specification/Repair | Next Step |
|---|---|---|

| | | |
|---|---|---|
| Check the sea water pressure signal and sensor supply +24-VDC wires for an open.<br><br>NOTE: An open has been detected by the remote input/output unit if a false alarm has occurred.<br><br>• Place one test lead on the sea water pressure signal pin at the C4 connector.<br>• Place the other test lead on the sea water pressure signal pin at the C11 connector.<br>• Place one test lead on the sea water pressure sensor supply +24-VDC pin at the C4 connector.<br>• Place the other test lead on the sea water pressure sensor supply +24-VDC pin at the C11 connector.<br>• Place one test lead on the sea water pressure signal pin at the C11 connector.<br>• Place the other test lead on the sea water pressure signal pin at the sensor connector.<br>• Place one test lead on the sea water pressure sensor supply +24-VDC pin at the C11 connector.<br>• Place the other test lead on the sea water pressure sensor supply | Less than 10 ohms?<br><br>**YES** | **2B** |

| | | |
|---|---|---|
| +24-VDC pin at the sensor connector.<br><br>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification. | Less than 10 ohms?<br><br>**NO**<br><br>**Repair :**<br><br>Replace the wire. Refer to the OEM service manual. | **Repair complete** |

### STEP 2B. Check the sea water pressure signal and sensor supply +24-VDC wires for a wire-to-wire short.

**Conditions :**

- Disconnect the OEM harness at the C4 and C11 connectors.
- Disconnect the sea water pressure sensor connector.

| Action | Specification/Repair | Next Step |
| --- | --- | --- |
| Check the sea water pressure signal and sensor supply +24-VDC wires for a wire-to-wire short.<br>• Place one test lead on the sea water pressure signal pin at the C4 connector.<br>• Place the other test lead on all other pins at the C4 connector.<br>• Place one test lead on the sea water pressure sensor supply +24-VDC pin at the C4 connector.<br>• Place the other test lead on all other pins at the C4 connector.<br>• Place one test lead on the sea water pressure signal pin at the C11 connector.<br>• Place the other test lead on all other pins at the C11 connector.<br>• Place one test lead on the sea water pressure sensor supply +24-VDC pin at the C11 connector.<br>• Place the other test lead on all other pins at the C11 connector.<br>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification. | Less than 10 ohms?<br>**YES**<br>**Repair :**<br>Replace the wire. Refer to the OEM service manual. | **Repair complete** |
| | Less than 10 ohms?<br>**NO** | **2C** |

### STEP 2C. Check the sea water pressure signal and sensor supply +24-VDC wires for a short to ground.

**Conditions :**

- Disconnect the OEM harness at the C4 and C11 connectors.
- Disconnect the sea water pressure sensor connector.

| Action | Specification/Repair | Next Step |
|---|---|---|
| Check the sea water pressure signal and sensor supply +24-VDC wires for a short to ground.<br>• Place one test lead on the sea water pressure signal pin at the C4 connector.<br>• Place the other test lead on engine ground.<br>• Place one test lead on the sensor supply +24-VDC pin at the C11 connector.<br>• Place the other test lead on engine ground.<br>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification. | Less than 10 ohms?<br>**YES**<br>**Repair :**<br>Replace the wire. Refer to the OEM service manual. | **Repair complete** |
| | Less than 10 ohms?<br>**NO** | **2D** |

### STEP 2D. Check the sea water pressure sensor supply +24-VDC wire for voltage.

**Conditions :**

- Disconnect the sea water pressure sensor connector.

| Action | Specification/Repair | Next Step |
|---|---|---|

| | | |
|---|---|---|
| Check the sea water pressure sensor supply +24-VDC wire for voltage.<br>• Place one test lead on the sea water pressure sensor supply +24-VDC pin at the sensor connector.<br>• Place the other test lead on engine ground.<br>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification. | +24-VDC?<br>**YES**<br>**Repair :**<br>Replace the sea water pressure sensor. Refer to the OEM service manual. | **Repair complete** |
| | +24-VDC?<br>**NO**<br>**Repair :**<br>Replace the remote input/output unit. Contact a Cummins® Authorized Repair Location. | **Repair complete** |

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